

11	A	Using $pV = nRT$, $n = \frac{pV}{RT}$ Tap closed: Total amount of gas in containers X and Y $n_{\text{total}} = n_X + n_Y$ $= \frac{p_X V_X}{RT} + \frac{p_Y V_Y}{RT}$ $= \frac{100 \times 4}{RT} + \frac{50 \times 6}{RT}$ $= \frac{400}{RT} + \frac{300}{RT}$ $= \frac{700}{RT} \text{ --- (1)}$ Tap open: let final pressure of both containers be p' , $n_{\text{total}} = n'_X + n'_Y$ $= \frac{4p'}{RT} + \frac{6p'}{RT}$ $= \frac{10p'}{RT} \text{ --- (2)}$ Equating (1) and (2),

Qn	Ans	Solution
		$\frac{10p'}{RT} = \frac{700}{RT}$ $p' = 70 \text{ Pa}$
12	A	By conservation of energy the loss of heat energy store by glass stopper is